

Plant Hardiness Zones

A plant's ability to withstand relative low and high temperatures is referred to as cold-hardiness and heat tolerance, respectively. To help gardeners select plants based on these factors, the US Department of Agriculture has divided the United States into hardiness zones. Although these zones serve as a reliable guide, they are not the only indicator of the possible success of a plant. Other factors, such as soil type, fertility, and sun exposure, can affect it. All plants in this encyclopedia have been assigned zone ratings determined by the lowest temperature the plant is likely to withstand.

ZONE	TEMPERATURE RANGE	CATEGORY	DEFINITION
1	below -50°F (-46°C)	Severest winter	Only the hardiest plants are able to withstand winter in this zone. Others will require protection from cold temperatures.
2	-50° to -40°F (-46° to -40°C)	Severe winter	Only very hardy plants will survive winter in this zone. Most plants will require protection from cold temperatures.
3	-40° to -30°F (-40° to -34°C)	Severe winter	Evergreens, grasses, and hardy perennial plants that can tolerate cold winters will grow in this zone.
4	-30° to -20°F (-34° to -29°C)	Severe winter	Perennials in this zone, along with numerous evergreens and grasses, tolerate the very cold winters in this zone.
5	-20° to -10°F (-29° to -23°C)	Cold winter	A great variety of perennials, deciduous trees, evergreens, grasses, and annuals can tolerate the cold winters in this zone.
6	-10° to 0°F (-23° to -18°C)	Cold winter	Cold winters of relatively short duration in this zone means a wide variety of nontropical plants will survive here.
7	0° to 10°F (-18° to -12°C)	Average winter	Where rainfall is sufficient, the relatively mild winters and long summers here support a wide variety of garden plants and trees.
8	10° to 20°F (-12° to -7°C)	Mild winter	Mild winters and long, hot summers support a wide variety of plants here, but high humidity may be a problem for some species.
9	20° to 30°F (-7° to -1°C)	Mild winter	Mild, subtropical conditions allow plants from all categories to grow here, unless the plant needs a cold period to thrive.
10	30° to 40°F (-1° to 4°C)	Subtropical	Tropical plants grow well here. Some popular species from temperate regions may find it too warm.
11-15	Above 40°F (Above 4°C)	Tropical	Tropical and subtropical plants dominate gardens in this zone. Temperate species may adapt with special care.

ZONE NOTE

The zones presented in this book are offered as approximate guides and should not be considered as absolute. Practicing horticultural techniques such as mulching and winter protection will alter a plant's interaction with its environment and may enable it to survive beyond the zones given in this or other publications.

Page headings

Headings reflect the way in which each plant group is subdivided—usually by size and main season of interest. (See also Size categories, left.)

Color tabs

These indicate a change of color within the size group for each season.

Plant portraits

Color photographs assist in the identification and selection of plants.

Tags

Color-coded tabs make it easy to find each plant group.

Plant names

The botanical name is given for each plant, and where appropriate, common names are listed in brackets.

Captions

Captions describe the plants in detail and draw attention to any special uses.



The Plant Dictionary

The Plant Dictionary contains entries for every genus in the Encyclopedia and includes an additional 4,000 recommended plants to those featured in the Plant Catalog. It also functions as an index to the Plant Catalog.

Genus names

The genus name is followed by common names and synonyms, where appropriate, and family names.

Genus entries

A concise introduction covers the distinctive characteristics and hardiness range of plants in the genus, as well as advice on siting, cultivation, propagation, and, if relevant, pruning, pests and diseases, and toxicity.

Plant names

Botanical names, synonyms, and common names are given as appropriate. The genus name is abbreviated; specific epithets (eg. *nobile*) are abbreviated only if previously given in full.

Plant descriptions

Key characteristics of the plant are described. Hardiness and cultivation needs are included only if specific to the plant. Cultivar entries run on from the species entry, with the binomial omitted.

Illustrated plants

Descriptions for illustrated plants appear in the Plant Catalog, unless part of a feature panel (see below left).

Cross-references

Synonym cross-references are listed alphabetically.

RHEUM

Rhubarb

POLYGONACEAE

Genus of rhizomatous, often tough or woody, perennials, grown for their foliage and overall appearance. Includes the edible rhubarb and various ornamental plants. Some species are extremely large and require plenty of space. Fully hardy. Prefers sun or partial shade and deep, rich, well-drained soil. Propagate by division in spring or by seed in fall ☉ Leaves may cause severe discomfort if ingested.

R. 'Ace of Hearts', syn. *R. 'Ace of Spades'*. Clump-forming perennial. H to 4ft (1.2m), S 3ft (90cm). Has elongated, heart-shaped, dark green leaves, to 14in (35cm) long, red-veined above, purple-veined beneath. Bears long panicles of numerous, tiny, star-shaped, very pale pink to white flowers in mid- to late summer. Z4-9
R. 'Ace of Spades'. See *R. 'Ace of Hearts'*.
R. palmatum. Clump-forming perennial. H to 8ft (2.5m), S to 6ft (1.8m). Has 2-2½ft (60-75cm) long, rounded, 5-lobed, mid-green leaves. In early summer, bears broad panicles of small, creamy-green to deep red flowers. Z5-9.
'Atrosanguineum' illus. p.439.

Rhipsalidopsis gaertneri. See *Hatiora gaertneri*.

Rhipsalidopsis rosea. See *Hatiora rosea*.